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2 September 1976
SUPERSEDING
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MILITARY SPECIFICATION

GENERATORS, EXOTHERMIC ATMOSPHERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers atmospheric generators designed to produce an exothermic gas for purging, blanketing and metallurgical processes.

1.2 Classification. The generator size shall be based on cubic feet per hour (CFH) output of atmosphere. The generators shall be of the sizes listed in Table I and the size required shall be as specified (see 6.2.1).

TABLE I. Generator Output Capacity.

Size	Cubic Ft. per Hour	Size	Cubic Ft. per Hour
1	400	7	6000
2	800	8	8000
3	1500	9	10000
4	3000	10	15000
5	4000	11	20000
6	5000	12	30000

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 3424

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SPECIFICATIONS

Military

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

JOINT INDUSTRIAL COUNCIL (JIC)

EGP-1 Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to Joint Industrial Council, 2139 Wisconsin Avenue, NW, Washington, DC 20007.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulation, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

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NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete generator for first article inspection and approval (see 4.2 and 6.4).

3.2 Design. Generators provided in accordance with this specification shall be new and of the manufacturers current design. The generators shall be designed to produce an exothermic atmosphere by partially burning a predetermined quantity of hydrocarbon gas with air in a refractory insulated combustion chamber. The atmosphere shall be produced by the reaction of the gas-air mixture ranging from 55 to 90 percent of perfect combustion, at a temperature of from 2000 to 2600°F. The combustion of the hydrocarbon constituent shall produce a gas that is basically hydrogen and carbon monoxide through the heat generated by burning part of the gas to carbon dioxide and water vapor. The water vapor from the partial combustion shall be condensed out of the reacted gas and separated from the condenser by means of a water trap. The generator shall consist of a gas-air mixing system with mixing pump or blower, refractory,insulated combustion chamber, one or more burners, condenser system and water trap, flow meters and all necessary gas proportioning regulators, pressure relief and control valves, safety shut-off valves and switches. All components shall be piped, wired, and mounted on a common base to form a packaged unit. Operation of the generator shall provide for selecting any gas-air ratio required to produce an atmosphere with an analysis in the range specified herein. The air-gas ratio setting to produce the analysis of atmosphere required shall be made after the combustion chamber has heated to a temperature of 2000°F. After the air-gas ratio setting is made, the generator shall operate automatically and hold this ratio constant regardless of the rate of flow or variable atmosphere demand placed on the generator. Unless otherwise specified, the generator shall be designed to operate on natural gas having a heat content of 1000 BTU/ cubic foot and a supply pressure of 6-8 inches water column. When required, the generator shall be designed to operate on butane, propane, or manufactured gas having a specific analysis and specific gravity, as specified (see 6.2.1).

* 3.2.1 Safety devices. The generator shall be provided with a safety shutoff valve in the gas supply piping arranged to close in case of low gas pressure, high gas pressure, loss of air supply, or power failure.

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A low pressure switch and a high pressure switch shall be provided in the gas supply piping that will close the safety shutoff valve, and shut off the air supply or mechanical mixer, if the gas pressure varies more than plus or minus 10 percent of operating pressure. Generators with a separate combustion-air supply shall be provided with a low air pressure switch in the air supply piping to close the safety shutoff valve and shut off the power source to the blower when the air supply pressure is reduced more than 10 percent of operating pressure. A manual cock or valve shall be provided on the down stream side of the safety shutoff valve, with a tap in between, to permit periodic checking of the safety valve. Flow indicators, meters, or differential pressure devices on the gas and air supply piping, or a test burner with suitable flashback protection, shall be provided in the air-gas mixture line to aid in checking the air-gas ratio. If the generator employs a premix air-gas ratio system, the system shall be provided with automatic fire check protection, which, when actuated, shall close the safety shutoff valve in the gas line and shut off the mixing system. The generator shall be provided with a manual shutoff valve in the gas supply line, located upstream from the safety shutoff valve. This valve shall be readily accessible for emergency and normal shutdown. An alarm bell and red warning light shall be provided that will give both an audible and visual signal in the event the generator is shut down by any safety device. If the generator is shut down by any safety device, it shall remain down until manually restarted. The safety system of each generator shall conform to FM "Handbook of Industrial Loss Prevention". Except as otherwise specified by the procuring activity, the generator shall comply with the standards promulgated under OSHA applicable to the generator itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.2.3 Material. All material used in the fabrication of the generator and related equipment shall be new and of the quality necessary to produce a generator to meet the requirements described herein. Materials that are exposed to the internal environment of the generator chambers shall be compatible with the atmosphere and temperature conditions they will encounter.

3.2.3 Lubrication. Means shall be provided for lubrication of all sliding and rotating parts requiring lubrication, except motors or pumps with non-additive pack seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material, or equipment. Lubrication points shall be readily accessible, and identified in the operator's manual with type of lubricant and instructions for application.

3.2.4 Accessibility. All parts subject to wear, breakage, or distortion, and all parts which require periodic maintenance shall be accessible for adjustment and replacement, as applicable.

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3.2.5 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or equipment.

3.3 Construction. The generator shall be constructed so that when installed and connected to the required supply sources, it can be used for any purpose described herein. All parts and accessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where required in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the generator shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

* 3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.4 Electrical equipment. The generator shall be designed and furnished complete with all electrical motors, contactors, controls, relays, transformers, signal devices, lights and switches pre-wired where possible for operation upon connection to source of power. All power actuated, manually adjusted and automatic controls shall operate on 115 volt, 60

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cycle power supply. Generators that operate on 230/460 volts shall be provided with a stepdown transformer for the 115 volt system. A single disconnecting means shall disconnect all ungrounded conductors, and an identified terminal shall be provided for connecting the size grounding conductor specified in Table 15-1 of JIC EGP-1. All electrical equipment shall conform to JIC Standard EGP-1.

* 3.4.1 Motors. The electric motors shall be rated for continuous duty and shall be equipped with ball or roller bearings. Motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volts, 3 phase, 60 Hz and unless otherwise specified (see 6.2.1), initially wired to operate on 230 volts.

* 3.4.2 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

* 3.5 Performance. The generator shall produce atmosphere at a pressure ranging from 0.5 to 1.5 PSIG. When measured at the discharge side of the condenser at a temperature of 70 degrees F (± 10 degrees F) and operating at 100 percent of rated capacity, the generator output of atmosphere shall be not less than the cubic feet per hour specified for the size generator. The dew point of the produced atmosphere shall be not more than 10°F higher than the temperature of the cooling water at the inlet side of the condenser. The generator shall automatically maintain the air-gas ratio at any set value within the minimum and maximum output capacity. The generator operating with an air-gas ratio mixture of 6 parts air to 1 part natural gas shall produce an atmosphere with the following analysis:*

Carbon dioxide	4.0% ± 1%
Carbon monoxide	10.0% ± 1%
Hydrogen	18.0% ± 5%
Methane	0.12% ± .1%
Nitrogen	Balance

The generator operating with an air-gas ratio mixture of 9 parts air to 1 part natural gas shall produce an atmosphere with the follcwing analysis:*

Carbon dioxide	10.0% ± 1%
Carbon monoxide	3.0% ± .5%
Hydrogen	2.5% ± .5%
Nitrogen	Balance

*NOTE: The specific analysis of produced gas will depend on the specific analysis of the fuel gas input.

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3.6 Components and systems.

3.6.1 Base. The base shall be fabricated of angle or channel iron. The entire unit shall be reinforced with steel members and shapes, and welded to form a structurally sound unit that may be skidded or moved without damage. If legs are used, they shall be provided with bearing plates.

3.6.2 Combustion chamber. The combustion chamber shell of generators with 1500 CFH output and larger shall be of double wall construction and water jacket cooled. The shell shall be fabricated of 12 gauge steel sheet or thicker. End plates shall be gasketed and bolted. Water cooled generators shall be provided with an automatic water temperature control system and cooling water failure protection. The protective system shall function to shut the generator down in case of cooling water failure.

3.6.2.1 Chamber lining. The combustion chamber lining shall consist of refractory brick with a temperature rating not less than 2800°F backed up with insulating slab or preformed insulation. Catalyst, if used, shall be nickel nitrate impregnated refractory that will withstand a temperature of not less than 2600°F and give sufficient constant area to produce full reaction.

3.6.3 Heat exchanger. The generator shall be provided with a heat exchanger and condenser system capable of cooling the produced atmosphere to a temperature of 90°F or less. The system shall automatically remove moisture from the atmosphere to a dew point within 10 degrees F of the inlet cooling water temperature. When required, the generator shall be provided with refrigeration and dryer systems to further reduce the dew point of the atmosphere to the temperature specified (see 6.2.1).

* 3.6.4 Air-gas mixer and burner system. Each generator shall be provided with one or more burners, a gas-air mixing system and all necessary controls to proportion, mix, and process the mixture through the generator system. The mixing system shall automatically maintain the air-gas ratio at any set value within the minimum and maximum output capacity of the generator. The system shall be provided with two flow meters to indicate in cubic feet per hour the air and gas input to the generator. The incoming air shall be filtered. A spark igniter pilot system shall be provided for lighting the burner. All controls shall be mounted and wired to conform to JIC Standard EGP-1.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

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* 3.8 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate identifying the generator in accordance with MIL-STD-130 shall be securely attached to each generator. The nameplate shall contain the information listed below. If the generator is a special model, the model designation shall include the model of the basic standard generator and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Type of input gas and requirement (CFH)
Output capacity (CFH)
Date of manufacture
Contract or Order Number
National Stock Number or Plant Equipment Code
U S

* 3.9 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the generator and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

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* 4.3 Quality conformance inspection. Each generator shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 and the inspection in 4.6. Failure of the generator to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The generator and equipment shall be examined to determine compliance with the requirements of this specification.

4.5 Tests.

4.5.1 Operational and safety test. The generator shall be prepared for and operated in accordance with the manufacturer's handbook of instructions until the refractory and insulation has cured. With the generator operating at an atmosphere producing temperature between 2000 and 2600 degrees F, each safety control device and system on the generator shall be tested. Each system shall be tested by simulating a failure that normally would cause the device to operate. Each system shall perform its intended function.

4.5.2 Performance test. The generator shall be operated for not less than 3 hours at a temperature above 2000°F, and an atmosphere composition analysis and a flow rate analysis shall be performed. Not less than 3 series of tests shall be performed with an air-gas ratio mixture of 6 parts air to 1 part gas, and again with an air-gas ratio mixture of 9 parts air to 1 part gas. Each test shall be performed at intervals of not less than 1/2 hour. The atmosphere composition, dew point, flow rate, and discharge pressure shall conform to the requirements of 3.5. The capability of the generator to automatically maintain a constant air-gas ratio at various flow rates and variable atmosphere demands shall be checked and shall conform to the requirements specified herein.

4.5.3 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.2.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with the applicable requirements of MIL-F-3296. The required level of preservation and packing, and any special marking requirements shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. Generators covered by this specification are intended to produce an exothermic type of atmosphere for use in heat treating furnaces to prevent surface oxidation of ferrous materials, where decarburization is not a factor. The atmosphere may be used for such processes as bright annealing, bright normalizing, bright tempering of steel, bright copper or silver brazing of steel, silver brazing of non-ferrous mixtures in powder metullurgy and sintering of carbon powders.

* 6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identifed to each of the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. First article test, if required (see 3.1 and 6.4).
d. Gas analysis and specific gravity, if different (see 3.2).
e. When safety and health requirements are different; specify and fully describe (see 3.2.1 and 6.5).
f. Painting, if different (see 3.3.5).
g. Voltage for initial wiring of motor, if different (see 3.4.1).
h. Electromagnetic interference control, if required (see 3.4.2).
i. Dew point of produced atmosphere, if different (see 3.6.3).
j. Repair parts, if required (see 3.7).
k. Nameplate, if different (see 3.8).
l. First article inspection, if different (see 4.2).
m. Quality conformance inspection, if different (see 4.3).
n. Required level of preservation, packing and special marking requirements (see 5.1).

* 6.3 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

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* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Mercury prohibited. When the use of mercury or its compounds in the generator and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

* 6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
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Review Activities:
Army - WC
Navy - SH
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User Activities:
Navy - MC, AS, OS
Army - SM, MU, EL, WV

Preparing Activity:
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